

Build Your First Add-on for Mozilla Firefox in 30 Minutes

Mozilla Firefox is a fast, multi-platform Web browser with an extremely large user base. Firefox is made highly customisable by the use of thousands of add-ons, but there is scope for more. This tutorial demonstrates how to build a Firefox add-on in a very short time.

This article presents the basics required to create your first Mozilla Firefox add-on. Mozilla Firefox is considered one of the best open source Web browsers developed for Windows, OS X and Linux, by the Mozilla Foundation.

Its key features include:

- Tabbed browsing
- Spell-checking
- Incremental find
- Live bookmarking
- Smart bookmarks
- Location-aware browsing (geo-location based on Google services)
- ...and many more

Additionally, Firefox allows third-party developers to add functions through extensions or add-ons. It also provides an environment for Web programmers, where they can use dev tools, such as the DOM Inspector, the Error Console or add-ons such as Firebug.

What is an add-on?

An add-on is an installable enhancement for a particular application. It allows the user to add or augment extra application features, such as a theme or any third-party feature. Add-ons are more popular with Firefox because it retains a high degree of extensibility, giving individual users the freedom to add the features that they like.

Add-on technologies

The technology behind an add-on may consist of one or more of the following:

- XUL (XML user interface language): This is used to design and define the user interface and the interaction with the user. A majority of the legacy Mozilla add-ons are written in XUL. It is one of the best and oldest ways to create a Mozilla add-on.
- Mozilla Jetpack: This is an add-on SDK aimed at speeding up the development time and minimising the learning period. It is also used to develop add-ons that do not require users to restart the system.
- JavaScript: This is the primary language of the Mozilla

browser and is used for writing the logic for add-ons.

- DOM (Document Object Model): This is used to access or edit the HTML document that is loaded or to change the XUL interface in real-time.
- XPI (Cross-Platform Installer), XPConnect and XPCOM (Cross-Platform Component Object Model).

That's the theory part; now let us start with the practical aspects.

Creating an add-on

We are now going to develop a simple yet powerful add-on for Mozilla Firefox. Let us call it the DuckDuckGo search add-on, which will let you search on the DuckDuckGo search engine through the Mozilla Firefox browser, and return the search results in the browser.

